3.3.1 Number of research papers published per teacher in the journals notified on UGC Care list during the last five years

2023

List of paper publications

S.NO	Title of the paper	Name of the author/s	Name of the journal	Page No
1	Plasmonics and its applications	**Durga Pallavi Arumilli, Harika Garsikuti, Venkata Sita Mahalakshmi Gandrotula, Eswara Sri Rama sita Swapnika Ganta, Dr.v.Bhaskara Raju**	International Journal Of Novel Research and Development	3
2.	3D and 4 D Printing In Pharmaceutical Technology	**Arumilli Durga Pallavi, Kuna Lakshmi Prasanna, Dake Sushma, Noolu Likhitha Priya, Dr.v.Bhaskara Raju**	International Journal Of Novel Research and Development	4
3.	Nanosponges- An Innovative and Novel drug delivery system for drug targeting	**A.Durga Pallavi, M.Ajay Ratnam, P.Esther,P.Rambabu, Padala Narayana Raju**	International Journal For Research Trends And Innovation	5
4	An Overview Of Green Chemistry	**Sharmila M, Gayathri Devi NBS, Sharmila N, Saiteja PBSN**	Journal Of Pharma Insights and Research	6
5	Comprehensive Analysis On Compounds Derived From Schiffs Bases	**Navya Sree Devarapalli, Sindhu Sree Cholla, Vani Hymavathi Dayana, Ratna Deepika Pasumarthi, Kavya Sri Padala, Chudamani Naga Sravani Parasa, Antro Jennie Xavier, Bhaskara Raju Vatchavai.**	Journal Of Pharma Insights and Research	7
6	A Comparative review on Regulatory Frame work of Paediatric Drugs in the US and EU-Challenges and Recommendations	**V.Sri Sowmya, Y.B.Manjulatha, J.N.B.Indusekhar, V.Bhaskara Raju.**	International Journal Of Drug Regulatory Affairs.	8
7	Anticataract Potential of Piper Betel Leaf Ethanol extract on Dexamethasone induced cataract.	**Vijayambica Durga B, Priyanka Metla, Sneha Moka, Varaprasad Midde.**	Journal Of Pharma Insights and Research	9
8	Insilico studies of compounds isolated from methanol extract of elehantopus scaber by GCMS	**Antro Jennie X,Jenilia Bejads X, Rajesh M, Jobin Rex R, Sharmila .M,Bindu D,BhaskarRaju.V,**	Journal Of Global Trends In	11

	against protein PPAR-r for its hypoglycemic activity.	**Sathpathy.DK.Srinivas .M.**	Pharmaceutical Sciences.	
9	Impose of KNDy/GnRH neural circuit in PCOS,ageing,Cancer and Alzheimers disease :star actions in prevention of neuro endocrine dysfunction	**Siva Prasad panda,**Adarsh kesharwani,Gaurav deep singh,DSNBK Prasanth, Bhaskhar raju vatchavai,P.V.Kamala Kumari, Sunil Kumar panda,Sarada prasanna malik	Ageing research reviews	12
10	Evaluation of anti inflammatory activity of Millingtonia Hortensis leaf extract	**Krishnaveni Manubolu,**Ravisha peeriga,Mohan Gandhi Bonthu,Bhaskara raju vatchavai,Nirmala Korukola,Edward raju gope,Devi Kadirvel	International journal of chemical and biochemical sciences	13
11	A bibliometric analysis of investigations on black pepper published from 1978 to 2023	**Mohan Gandhi Bonthu,**Raveesha peeriga,Krishnaveni Manubolu, Bhaskara raju vatchavai,Nirmala korukola,Edward raju gope	International journal of chemical and biochemical sciences	14
12	Development and validation of an LC-MS/MS Bioanalytical approach for the quantitative analysis of agomelatine in human plasma	**Vijayalakshmi marella,** Mohan Gandhi bonthu, Suneetha achanta, Bhaskara raju vatchavai, Pavani gayathri allure,Srala Nekkalapudi	European chemical bulleitein	15
13	A review article on chalcones as anticancer agents	**Bindu dwarampudi,** Sharmila Manukonda,Vijaya Ambica Durga Bonthu, Dr,Bhaskar raju vatchavai.	Journal of cardiovascular disease research	16
14	A review on high performance liquid chromatography	**SS Malleswara Sharma sonti**	Journal of pharma insights and research	17-18
15	A new stability indicating RP-HPLC method for simultaneous estimation of Voxilaprevir,sofosbuvir,velpatasvir in bulk and pharmaceutical dosage forms	**T.Yamini kumari,**A.srinivasarao,A.K.M.Pawar	International journal of pharmaceutical research and applications	19
16	A Novel RP-HPLC Method development and validation for the simultaneous estimation of Drotaverine hydrochloride and mefenamic acid in pure drugs and formulation	**K.S.Sumanth,** V.Bhaskar raju,K.Kalyani,B.Mohan Gandhi,A.Pavani Gayathri,A.Swapna,P.Brahma teja,K.Ramanjaneyulu	Gradiva Review journal	20

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PLASMONICS AND ITS APPLICATIONS

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Abstract: In recent years, plasmonics has emerged as a revolutionary field of study with immense potential for various applications. This article explores the fascinating world of plasmonics and delves into its wide-ranging applications. Plasmonics is the study of the interaction between light and free electrons in metallic nanostructures, giving rise to surface plasmon polaritons (SPPs). These SPPs have unique properties that allow for the manipulation and control of light at the nanoscale. By harnessing these properties, plasmonics holds promise in fields such as sensing and biomedical applications.

IndexTerms **– Plasmonics, Applications, Surface plasmons**

INTRODUCTION

Plasmonics is an entirely novel area of expertise that has sprouted owing to noble metal nanostructures capacity to regulate illumination at the nanoscale. These nanostructures have broad range of usage in chemical and biological sensing, medicines, photothermal therapeutics, food sciences, environmental sciences and solar energy. Nowadays, It is feasible to produce metal particles with heterogeneous compositions, which comprise plasmonic materials, and control their dimensions, form, besides arrangement. [1]First discovered as surface plasmon polaritons in the 1950s marked plasmonics' inception.Plasmonics, also called as nanoplasmonics, is a rather young area of study within the domains in nanophotonics along with nanooptics. The study of electron oscillations in metallic nanostructures and nanoparticles (NPs) is the core focus of plasmonics.[2]When electromagnetic radiation is applied to a metal surface, conduction electrons on the surface oscillate coherently, creating surface plasmons (SPs) at the metal-dielectric contact.The field of plasmonics is likewise quite busy because of recent developments in nanofabrication techniques.Metal nanostructures made up of nanoparticles (NPs), nanoholes, and other elements with carefully regulated sizes, shapes, and/or spacings have been made possible by these techniques.[3]A major advancement in our comprehension of the science underpinning plasmonics is made possible by the combination of nanometer-scale production processes and increasingly complex numerical modeling capabilities. [4] Surface plasmon resonance is the collective oscillation of free electrons in a plasmonic nanoparticle at a resonant frequency caused by an alternating electric field under light irradiation. Calculations and tests have demonstrated that theparticle shape, which controls the distribution and polarisation of free electrons on the surface, has an impact on the resonance's frequency and amplitude. Thus, the most effective way to modify and fine-tune an optical resonance property of a plasmonic nanoparticle is to change its shape.[5]The field of optical sensors has experienced tremendous growth and widespread applications in recent years. Surface plasmon resonance has established itself as an immensely acclaimed and influential optical sensing tool with quintessential applications in life sciences, environmental monitoring, clinical diagnosis, pharmaceutical developments, and ensuring food safety.Surface plasmon resonance (SPR)-based optical sensorsHave become one of the most frequently used types of sensors. They are extremely useful and adaptable, offeringA broad spectrum of commercial and research opportunities in the chemical and biological sciences. [6]

This review article includes various applications like Detection of various viruses, Detection of antibodies, Biosensors, Plasmonic photothermal monitoring, Monitoring haemoglobin levels in blood, DNA based plasmonic structures, In pharmaceutical analysis, Diagnosis of dengue virus, Detection of cancer, Detection of cyanide, Plasmonic Nanostructures in Sensor

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3D AND 4D PRINTING IN PHARMACEUTICAL TECHNOLOGY

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Abstract: 3D printing enables the creation of customized medication, tailored to an individual's specific requirements. This technology allows for precise dosage adjustments, combination therapies, and the incorporation of multiple active ingredients into a single tablet. 4D printing has emerged as a cutting-edge technology in the pharmaceutical field. 4D printing takes the concept of 3D printing a step further by incorporating materials that can change their shape or properties over time in response to external stimuli, such as temperature or moisture. This dynamic capability opens up endless possibilities in drug delivery and tissue engineering. The integration of 4D printing with drug delivery systems enables the development of smart drug carriers that can respond to specific stimuli in the body. For example, a 4D-printed capsule can release its contents at a predetermined location or time, optimizing drug efficacy. This technology has the potential to revolutionize the treatment of chronic diseases, where precise drug delivery is crucial. Furthermore, 4D printing holds great promise in tissue engineering, where the goal is to fabricate functional human tissues and organs. By utilizing materials that can transform their shape or properties, 4D printing allows for the creation of complex, self-assembling structures that mimic natural tissues. This advancement could transform the field of regenerative medicine, offering new possibilities for organ transplantation and personalized tissue implants.

***IndexTerms*– 3d&4d, Material, Techniques, Applications**

I. INTRODUCTION

In many industries, three-dimensional printing (3DP) is becoming the new standard for modern production tools. The way that products are laid out and manufactured is being drastically altered by this additive device. By using a 3-D computer model, customized products can be generated automatically layer by layer. 3DP seeks to offer customized tablets based on the needs of the patient. It considers the pharmacogenomics along with knowledge on their lifestyle, food, and environment. In 1986, Charles Hull became the first person to describe 3DP, a unique technology. Producing personalized pharmaceutical drug products is expedited by computer-based drug layout in conjunction with 3DP technology. The polypill idea in particular is a wonderful example of how 3DP technology has an embryonic in-person dose shape concept. The FDA in the United States approved Levetiracetam, or Spritam, as the main oral drug prescribed to treat epileptic seizures in 2015. Pharmaceutical manufacturing uses a variety of 3DP technologies, including Fused Deposition Modelling (FDM), Stereo lithography (SLA), Direct Energy Deposition (DED), Selective Laser Sintering (SLA), Thermal Injection Printing (TIP Printing), and Power Bed Injection Packaging. Growing interest in pharmaceutical preparation enhancement is being tapped into by 3DP as a potent way to address a few issues with traditional pharmaceutical unit operations. The standard operations of milling, coupling, granulation, and compression in a production unit may yield distinct characteristics for the end product when it comes to drug burdening, medicine delivery, drug stability, and pharmacological dosage form stability. Utilizing endure inject printing, the first wave of 3DP pharmaceuticals was created. Wu et al. developed the first drug delivery system with 3DP in 1996.[1]

Drug designers are constantly coming up with new concepts and refining their manufacturing techniques, materials knowledge, and manufacturing technology to provide superior dose patterns. When developing new drugs, active pharmaceutical ingredients (APIs) have a number of physicochemical and biological characteristics that must be considered and investigated. "Three-dimensional printing" is defined by the International Standard Organization (ISO) as "Fabrication of items through the deposition of a material utilizing a print head, nozzle, or other printer technology". Unlike the more widely used subtractive and formative manufacturing procedures, this approach is part of the additive manufacturing methodology. [2]

The creative process is endless. 4D printing is now a reality with the advent of the 3D printing era in medical and drug delivery. 4D printing adds a fourth dimension—time—to the 3D printing platform. In reaction to internal or external stimuli, products created via 4D printing frequently change their configuration over time (such as heat, p^H, temperature, or even water). The smart materials (feedstock) and the preplanned design of the good's fabrication process (referred to as smart design) may contribute to

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Nanosponges – An Innovative and Novel Drug Delivery System for drug targeting

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***Abstract*- An innovative and developing technology known as the nano sponges has an ability to target the drug to the desired site, where the drug is released in an exact and controlled manner. Nanosponges are spherical, porous, nanosized drug carriers with a three-dimensional structure created by crosslinking polymers. They can load wide variety of drugs, antibodies, proteins, vaccines and enzymes into their nanocavity and deliver them to the targeted site. These can be manufactured as oral, parenteral, topical, or inhalation dosage forms. Nanosponge's applications include cancer, carrying enzymes and biocatalysts, delivering oxygen, enhancing solubility, immobilizing enzymes, poison absorbents and COVID 19 treatment. This article has discussions on the use, preparation techniques, assessment criteria, recent patents and applications of nanosponges.**

***Keywords:* Nanosponges, solubility enhancement, oral, parenteral, topical, inhalation, drug targeting**

1. INTRODUCTION:

Targeting the medicament to the exact site in the human body and releasing that in a controlled manner are the two main problems medical researchers having from a long time. These problems can be overcome by nanosponges as a drug carrier. Nanosponges are a novel class of small particles whose structure look like sponge, which are used to treat many diseases and early experiments suggested that using this technology drugs can be targeted to the specific site for various types of cancers more effectively. [1] These particles are composed of holes which are nanometer-wide that can enclose a variety of medicinal ingredients and can entrap both lipophilic and hydrophilic compounds because of their inner hydrophilic and hydrophobic branching. [2] The backbone of nanosponges, which resemble a three-dimensional network, is a length of polyester combined with a crosslinking agent-containing solution. These crosslinking substances function as little hooks to connect various polymer components. They have a spherical colloidal nature and more solubilization capacity of drugs which have minimum solubility and hence improve their bioavailability. [3]

2. ADVANTAGES:

- This technique reduces negative effects while entrapping the substance.
- At temperatures as high as 130^{0}C, these compositions remain stable.
- The majority of vehicles and substances can use these kinds of combinations.
- Stability, elegance, and formulation flexibility have all improved by nanosponges. [4]
- The nanosponges assist in covering the unpleasant taste of drugs that are orally administered or through the buccal route. The negative effects of various substances can be minimized since they are getting entrapped within the nanosponges during formulation. [5]
- Nanosponges have stability up to 300°C temperature and pH ranges of one to eleven. Improved compatibility is seen when using various solvents and parts. They aid in modifying the pace at which medicine formulations release their active ingredients. [6]
- As medication is administered less often, patient compliance rises. They are easily scaled up, so they may be sold fast. Nanosponges drug moieties may be hydrophilic or lipophilic.
- Nanosponges are made using crosslinkers and other ingredients to shield the drug from hepatic first pass metabolism.
- Utilizing environmentally friendly solvents, inert hot gases, gentle heating, and adjusting strength, nanosponges may be regenerate.
- They can keep functioning for as long as 12 hours thanks to its extended-release feature. [7]
- Systems using nanosponges do not cause discomfort, mutagenesis, allergic reactions, or cytotoxicity. [8]

Sri Vasavi Institute of Pharmaceutical Sciences

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JOURNAL OF PHARMA INSIGHTS AND RESEARCH

REVIEW ARTICLE

An Over view of Green Chemistry

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Abstract: Over the past decade, green chemistry principles have demonstrated the potential to protect human health and the environment while reducing costs through the design of chemical products and processes. Products and methods designed according to principles that support life-cycle sustainability could form the basis of an environmentally benign chemical enterprise. While the Pollution Prevention Act of 1990 established a regulatory framework supporting green chemistry's development, its origins can be traced to earlier scientific efforts to minimize waste and hazards. Effective implementation of green chemistry principles requires overcoming challenges in systematically redesigning chemical synthesis routes. This review explores the historical context of green chemistry's growth, current applications across sub-disciplines like organic synthesis, and strategies to advance education and further innovation through alternative solvents and other design approaches. It examines progress in applying green chemistry concepts, remaining barriers, and promising future directions for more sustainable chemical technology.

Keywords: Green chemistry; Organic solvents; Environment; Sustainability; Pollution; Eco-friendly

1. Introduction

Green chemistry aims to reduce or eliminate the use of potentially hazardous substances in chemical products and their production processes. [1-3] It considers the full life cycle of a chemical product from its design, manufacture, use, and final disposal or recycling. The term 'green chemistry' emerged in the early 1990s to describe the design of chemical syntheses and technologies that are environmentally benign. [4-6] Significant early initiatives driving the development of green chemistry principles included the US Presidential Green Chemistry Initiative, launched in 1995, the formation of the Green Chemistry Institute in 1997, and the inaugural publication of the Royal Society of Chemistry's journal Green Chemistry in 1999. [7-9]

In the 1990s, Paul Anastas and John Warner proposed the 12 principles of green chemistry that provide a framework for synthesizing chemicals and designing production processes through safer, non-waste generating methods. This built upon earlier regulatory actions like the 1990 Pollution Prevention Act that established a foundation for reducing environmental impacts from the chemical industry. [10] Paul Anastas is widely recognized for pioneering the field of green chemistry through his research and promotion of its concepts. Nobel prizes in Chemistry have also recognized research advancing green chemistry goals, such as the awards to Knowles, Noyori, and Sharpless in 2001 for developments in asymmetric catalysis and to Chauvin, Grubbs, and Schrock in 2005 for advancements in metal-catalyzed olefin metathesis. [11]

2. Principles of Green Chemistry

The 12 main ideas of Green Chemistry are:

2.1. Prevent waste

Chemical syntheses should be designed to prevent the generation of byproducts and waste materials. This maximizes atom efficiency and mass yield. [12]

2.2. Design safer chemicals & products

The design of chemical products should reduce their potential to cause harm to living organisms and the environment. Toxicity concerns should be evaluated early in the design process through tools like quantitative structure-activity relationship modeling. [13]

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JOURNAL OF PHARMA INSIGHTS AND RESEARCH

REVIEW ARTICLE

Comprehensive Analysis on Compounds Derived from Schiff's Bases

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Abstract: Schiff bases, synthesized through the condensation of primary amines with carbonyl compounds, constitute a versatile class of compounds with broad applications. Their adaptable chemical structure enables the integration of diverse functional groups, leading to a spectrum of properties and utilities. Exhibiting significant characteristics in coordination chemistry, materials science, and organic synthesis, Schiff bases play a pivotal role in modern chemistry and technology. This article offers an encompassing overview of Schiff base synthesis, properties, and applications, emphasizing their pivotal role in contemporary chemistry and technology. Additionally, this review provides insights into recent advancements in Schiff base chemistry, including novel synthetic methodologies, their utility as ligands in metal complex catalysis, and their application in drug discovery and biomedical research. Overall, Schiff bases remain a dynamic field of research and innovation, offering promising prospects for the development of novel functional materials and compounds.

Keywords: Schiff bases; Imines; Azomethine; Hydrazones; Chitosan.

1. Introduction

Since their discovery by a German scientist in 1864, Schiff bases have emerged as condensation products of primary amines with carbonyl compounds, constituting a significant and versatile family of chemical compounds. Widely employed across various disciplines, including analytical, inorganic, and biological chemistry, Schiff bases derived from diverse heterocyclic compounds manifest an extensive array of biological actions. [1-4] These encompass analgesic, anti-inflammatory, anti-platelet, antioxidant, antiproliferative, antipyretic, cardio protective, antidepressant, antihypertensive, herbicidal, antiglycation, and cytotoxic properties, as well as antibacterial, antifungal, anti-mycobacterial, anti-anthrax, antiviral, anticancer, anti-protozoal, anti-parasitic, and anticonvulsant activities. [5-9]

Current research endeavors predominantly focus on synthesizing, characterizing, and elucidating the structure-activity relationships of Schiff bases. The nitrogen atom present in the azomethine (-NHN=PH-) or imine (-N=CH-) groups of these compounds garners particular attention due to its distinctive chemical and biological properties. Ashraf et al. demonstrated that the presence of a single pair of electrons in the sp2 hybridized orbital of the azomethine group's nitrogen profoundly influences its chemical and biological behavior. Schiff's seminal work in the 19th century marked the inception of imine production, catalyzing subsequent advancements in synthetic methodologies. [10-14] Various techniques have since been developed for synthesizing imines, with the traditional approach, as described by Schiff, involving the condensation of an amine and a carbonyl molecule through azeotropic distillation. [15, 16]

The synthesis of Schiff bases entails the electrophilic attack of the carbonyl carbon of aldehydes or ketones by the amine nitrogen from a primary amine, resulting in the formation of a carbinolamine (shown in Figure 1). Subsequent deprotonation of the carbinolamine's nitrogen facilitates the displacement of oxygen by the electrons from the N-H bond, yielding a compound characterized by an N-substituted imine and the expulsion of a water molecule. This N-substituted imine, also known as a Schiff base, exemplifies the versatility and significance of these compounds in contemporary chemistry and technology [17, 18]

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Review Article

Open Access

A Comparative Review on the Regulatory Framework of Pediatric Drugs in the US and EU: Challenges and Recommendations

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Abstract

Children and adults have different pharmacokinetic and pharmacodynamic profiles, so it is necessary to confirm the drug's effects in pediatrics. Drugs utilised in paediatric research They are frequently referred to as "therapeutic orphans" since they are challenging to develop and lack adequate data. Therefore, it becomes necessary to study the optimal dosages and formulations in various age groups of children. The significance of pediatric rules emerged from the fact that the physiological conditions of adults and children are different, and the same dosing regimen cannot be recommended for both. The child protection in research was recommended at the Belmont report (1979) first time. The necessity for written informed consent of the subject from a legally authorized representative of a child was described in the Declaration of Helsinki (1964). The scarcity of available patient populations, practical complexities of research, and minimal financial returns have hampered pharmaceutical investment in the clinical studies for children. More recently, pediatric policy in the US and Europe have instituted a system of obligations and incentives to stimulate investment in the pediatric drugs development. These initiatives, in conjunction with a more sophisticated process of drug discovery and development, resulting in significant advancements in the labelling of drugs for pediatric use. The present study reviews the regulation of pediatric drugs in the USA, EU and recent updates concerned to their regulations. It discusses the pediatric clinical study plans, Incentives, Timelines, challenges and possible recommendations. The challenges include ethical issues, clinical trial designs, type of formulation preparation, dosing, bioavailability and drug response measuring techniques.

Keywords: BPCA, PREA, Pediatric study plan, Pediatric investigation plan, Pediatric clinical trials

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Introduction:

Children need specific medicines and approaches to drug development. They have different pharmacokinetic and pharmacodynamics responses as compared to adults. These variations are mostly brought on by differences in body water and serum protein composition in the children. It is necessary to study the drug profile in different age groups of children and drugs available for adult may cause adverse effects in children. back then, and unfortunately still today, numerous medicines were given in an "off label" manner to children, even without a license or marketing authorisation. Inadequate testing can result in direct risk of under or overdosing and a delayed risk of long-term adverse effects in children. Important medicines must be tested in each pediatric sub-population. The USA was the first country which made the act for pediatric clinical trials as the "FDA Modernization Act" (1997). After that the "Best Pharmaceuticals for Children Act" (BPCA, 2002) and the "Pediatric Research Equity Act" (PREA, 2003) were incorporated in the U.S. regulatory framework. The first joint pediatric regulatory action was taken with the issuance of ICH E11 guideline. (1,2)

JOURNAL OF PHARMA INSIGHTS AND RESEARCH

RESEARCH ARTICLE

Anti-cataract Potential of *Piper Betle* Leaf Ethanol Extract on Dexamethasone-Induced Cataract

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Abstract:
The objective of this research was to assess the effectiveness of various ethanolic extracts derived from *Piper betle* leaves in combating cataracts, particularly in glucose-induced goat eye lenses used as in-vitro models. Cataracts, a prominent cause of vision impairment globally, manifest as the clouding of the eye lens. Dexamethasone, a frequently employed steroid, is recognized for its propensity to induce cataract formation. *Piper betle*, a traditional medicinal plant known for its anti-inflammatory properties, was investigated in this study. The research involved inducing cataracts in goat lenses using glucose. It was established that the phytoconstituents present in piper betle possessed notable antioxidant properties, contributing to their efficacy against cataracts. Examination of photographic evidence revealed that the application of various *Piper betle* leaf extracts delayed the onset of lens opacity. Additionally, the induced cataract lenses exhibited lower catalase levels compared to the normal control group, resulting in diminished opacity. *Piper betle*, also referred to as betel vine or Nagvalli, is widely cultivated in Asia and utilized for various medicinal purposes due to its astringent taste and affordability. The promising outcomes of this study underscore the potential of *Piper betle* as a therapeutic agent against dexamethasone-induced cataracts. The protective effects observed in the ethanolic extract of *Piper betle* leaves against dexamethasone-induced cataracts in isolated goat lenses suggest its anti-cataractogenic properties. These effects are likely mediated by the rich phytochemical composition of *Piper betle*, including polyphenols and flavonoids, which act as antioxidants, mitigating oxidative stress and preserving lens clarity. While this study provides valuable insights into the potential therapeutic benefits of *Piper betle* against cataracts, further research involving in vivo experiments and clinical trials is necessary to confirm these findings and assess the applicability of *Piper betle* as a cataract-preventive agent in humans.

Keywords: *Piper betle*; Anticataract activity; Dexamethasone-induced cataract; Phytochemical composition; Ocular disorders

1. Introduction

Betle leaf, alternatively known as betel vine or Nagvalli, holds significant medicinal value across Asia, with widespread cultivation in countries like Bangladesh, India, Sri Lanka, Malaysia, Thailand, Taiwan, and other Southeast Asian nations. Its diverse applications in traditional medicine, chewing practices, vaginal douching, mouthwash preparations, and remedies for skin ailments and coughs stem from its characteristic astringent taste. The leaf's affordability and abundance underscore its potential for further exploration in both the food and pharmaceutical industries. Moreover, betel leaf has long been a staple in traditional Indian customs, particularly in aiding digestion post-meals, with numerous varieties available worldwide, particularly concentrated in India, where it boasts 45 distinct types. Rich in tannins, sugar, diastases, and essential oils endowed with antiseptic properties, betel leaf presents a versatile botanical resource. [1-3]

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IN SILICO STUDIES OF COMPOUNDS ISOLATED FROM METHANOL EXTRACT OF *ELEPHANTOPUS SCABER* BY GCMS AGAINST PROTEIN PPAR- Γ FOR ITS HYPOGLYCEMIC ACTIVITY

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ABSTRACT

PPAR-γ is the abbreviation of peroxisome proliferator-activated receptor gamma, is also called as glitazone receptor, therapeutically used to fight against hyperglycemia associated with the metabolic syndrome and type -2 diabetes since the currently used PPAR- γ –agonist from the thiazolidine dione type shown serious side effects and hence making the discovery of novel ligands which are highly relevant, effective in normalization of blood glucose levels .The present study evaluated the binding of certain GCMS derived compounds of the methanol extract of the *Elephantopus scaber* leaves for its hypoglycemic activity of PPAR - γ. Docking scores were compared with their binding energy as well as affinity using the Schrodinger, 2015.1.The docking or glide score of the above seven compounds are -5.0,-8.7, -4.55,-3.7,-6.1,-5.8,-4.3(kcal/Mol) and the glide score were -33.43,-16.91,-29.21,-17.54,-29.79,-28.87 and -23.07 (kcal/Mol). The results of the study showed that out of the 9 isolated compounds, 7 of them may act as good agonist for PPAR- γ and the compounds can be re-designed for better hypoglycemic activity for synthesis.

INTRODUCTION

PPARs in structurally was identical to the steroid or thyroid hormone receptor and are stimulated in response to small lipophilic molecules. The PPARs exist in three subtypes; α, β, δ, and γ, each of which mediates the physiological actions of a large variety of FAs and FA derived molecules. Activated PPARs are also capable of transcriptional repression through DNA-independent protein-protein interactions with other transcription factors such as NFκB signal activators and transducers of transcription STAT-1 and AP-1 signaling (Olivera, 2007). Peroxisome proliferator-activated receptor gamma (PPAR-γ or PPARG), also called as glitazone receptor or NR1C3 (nuclear receptor subfamily 1, group C, member 3). It is a type II nuclear receptor that in humans is encoded by the PPARG gene. The molecular agonists of the nuclear receptor PPARγ are therapeutically used to fight against hyperglycemia associated with the metabolic syndrome and type II diabetes. Since, the

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Impose of KNDy/GnRH neural circuit in PCOS, ageing, cancer and Alzheimer's disease: StAR actions in prevention of neuroendocrine dysfunction

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ABSTRACT

The Kisspeptin1 (KISS1)/neurokinin B (NKB)/Dynorphin (Dyn) [KNDy] neurons in the hypothalamus regulate the reproduction stage in human beings and rodents. KNDy neurons co-expressed all KISS1, NKB, and Dyn peptides, and hence commonly regarded as KISS1 neurons. KNDy neurons contribute to the "GnRH pulse generator" and are implicated in the regulation of pulsatile GnRH release. The estradiol (E2)-estrogen receptor (ER) interactions over GnRH neurons in the hypothalamus cause nitric oxide (NO) discharge, in addition to presynaptic GABA and glutamate discharge from respective neurons. The released GABA and glutamate facilitate the activity of GnRH neurons via GABAA-R and AMPA/kainate-R. The KISS1 stimulates MAPK/ERK1/2 signaling and cause the release of Ca^{2+} from intracellular store, which contribute to neuroendocrine function, increase apoptosis and decrease cell proliferation and metastasis. The ageing in women deteriorates KISS1/KISS1R interaction in the hypothalamus which causes lower levels of GnRH. Because examining the human brain is so challenging, decades of clinical research have failed to find the causes of KNDy/GnRH dysfunction. The KISS1/KISS1R interactions in the brain have a neuroprotective effect against Alzheimer's disease (AD). These findings modulate the pathophysiological role of the KNDy/GnRH neural network in polycystic ovarian syndrome (PCOS) associated with ageing and, its protective role in cancer and AD. This review concludes with protecting effect of the steroid-derived acute regulatory enzyme (StAR) against neurotoxicity in the hippocampus, and hypothalamus, and these measures are fundamental for delaying ageing with PCOS. StAR could serve as novel diagnostic marker and therapeutic target for the most prevalent hormone-sensitive breast cancers (BCs).

1. Introduction

Kisspeptin (KISS) is a polypeptide encoded by the KISS-1 gene. The G-protein coupled receptor 54 (GPR54), additionally known as the KISS1 receptor (KISS1R), mediates the effects of kisspeptins, which are encoded by the KISS1 gene. The 145-amino-acid peptide encoded by the KISS1 gene is the first protein to be produced by the gene. This peptide is an arginine-phenylalanine (RF) amide. It is processed into smaller, biologically active peptides with names like kisspeptin-54 (KISS54), kisspeptin-14 (KISS14), kisspeptin-13 (KISS13), and kisspeptin-10 (KISS10), where the numbers represent the number of amino acids and KISS10 is the common C-terminal decapeptide sequence shared by all. Key for pulsatile GnRH secretion is a subset of neurons called kisspeptin (KISS)/neurokinin B (NKB)/Dynorphin (Dyn) neurons (KNDy neurons), which are situated in the hypothalamic arcuate nucleus. NKB acts primarily via its receptor (NK3R) to stimulate KNDy neurons.

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Evaluation of Anti-inflammatory Activity of *Millingtonia hortensis* Leaf Extract

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Abstract

In the present study "*Millingtonia hortensis*", a herbal aqueous extract of leaves are investigated to find anti-inflammatory activity by invitro evaluation. *Millingtonia hortensis* (BIGNONIACEAE) subjected for Invitro anti-inflammatory activity was evaluated by albumin protein denaturation method. *Millingtonia hortensis* leaf aqueous extract was prepared by cold maceration method. *Millingtonia hortensis* aqueous extract was screened and gave postive results for alkaloids, flavonoids, carbohydrates, phenols and other constituents. The extract gave more significant anti-inflammatory activity relative to reference standard drug sample ibuprofen. *Millingtonia hortensis* leaf aqueous extract showed noticeable anti-inflammatory activity and it indicates that it has potential application in the conditions of inflammatory conditions like arthritis, diabetes mellitus and cancer where natural anti-inflammatory agents are having beneficial action.

Keywords: *Millingtonia hortensis*, herbal extract, anti-inflammatory, albumin denaturation

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1. Introduction

This quickly growing tree sheds its flowers early in the morning and flowers at night. Hortensis means "grown in gardens," and the name *Millingtonia* is derived from the English botanist Thomas Millington. The tree is a gardener's favorite. Because its corky bark is used to process a subpar cork, it is also known as the "cork tree." The biological name of the tall, drought-resistant tree is *Millingtonia hortensis*, and it belongs to the bignoniaceae family.

The flowers have a rich, pleasant scent. Propagation using suckers and seeds. This crude drug is also known by the names Akas Nim, Nim chameli, betali nim, mini chameli, and karkku. It has an eternal lifespan. It's a tall tree, deciduous. It increases meter by meter. The leaves have multiple pinnate layers. Dried flower is a good lung tonic, used in cough diseases; bark is used to make yellow dye. Grows in tropical forests in central India [1]. Trees that are 8–25 meters tall. 40–100 cm in length, with elliptic, ovate, or ovate ot oblong leaflets that are 1.5–4 cm, glabrous, base rounded, oblique, margin entire, and apex acuminate; four or five lateral veins on the side of the midrid [2]. Cymose – paniculate inflorescence. Tiny, cupular, 2-4 mm, sino-late lobed calyx with slightly reflexed lobes. White corolla, tube 3-7 cm, diameter 2-3 mm. Glabrous ovary with many, four-rowed ovules.

Habitat low altitude slopes, usually located between 500 and 1200 meters at an elevation of 0 to 922 meters (0 to 3,025 feet). The medicinal uses include apoptosis and antiproliferation against RKO colon cancer [3]. It exhibits antimicrobial activity. Essential oil of flowers [4]. In Hispidulin and ethanoic extracts from Millingtonia hortensis and Oroxylum indicum (L) Kurz were chemically characterized and shown to have ant arboviral activity [5-8].

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A Bibliometric Analysis of Investigations on Black Pepper Published from 1978 to 2023

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Abstract

Spices have played a crucial role in human history, shaping cultures, trade routes, and culinary traditions for millennia. The diverse and fascinating world of spices, examining their origins, cultural significance, economic impact, and the multifaceted roles they play in both traditional and modern societies. Black pepper (Piper nigrum) stands as one of the most globally recognized and widely used spices, with a history rooted in ancient trade routes and culinary traditions. The multifaceted aspects of black pepper, encompasses its botanical characteristics, historical significance, culinary applications, medicinal properties, and its current standing in the global spice market. Therefore, this study aims to map investigations on black pepper using the bibliometric method. A bibliometric investigation was adopted through metadata planning with the keywords "Investigation AND black AND pepper" from Scopus Database (1978-2023). Metadata is stored in CSV and BibTex types. Furthermore, CSV format of Scopus metadata for analysis using the counting method on VOS viewer. Mapping results showed that the number of publications related to the black pepper experienced a minimum investigation, most occurring from 2007 to 2023. Most articles relating to Black pepper were published in the Food and Chemical Toxicology. The most prolific studies were conducted by Elangovan Kannan, Gunasekaran Vetrichelvi, Niranjali Devaraj is the most cited. Studies on black pepper need to be explored and most of the keywords with a fairly high density included phenols, hplc, antivenom. Meanwhile, the rarely investigated themes include piperaceae.

Keywords: Bibliographic, Citations, Black pepper, Vos viewer

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1. Introduction

Spices are plant-based ingredients used to flavour, colour, or preserve food. They are usually extracted from the seeds, bark, roots, fruits, or other parts of plants [1]. For millennia, they have been an essential component of human culinary customs and conventional medical procedures [2]. In addition to adding flavour and perfume to food, spices play an important role in the cultural and historical identities of many different cuisines worldwide [3]. Professionals in the food business can also benefit from using spices as a beneficial reference [4-5]. One of the most extensively used and traded spices in the world is black pepper (Piper nigrum) [6]. It is well known for both its strong flavour and adaptability in cooking [7]. Black pepper is said to have come from India's Malabar Coast and has long been a highly valued item in the spice trade [8]. Black pepper has been acknowledged for possible health advantages in addition to its culinary usage [9]. Because black pepper (Piper nigrum) has such cultural, gastronomic, and maybe health-related significance, it has been the focus of several scientific studies. Scholars have investigated several facets of black pepper, encompassing its chemical constitution, growing methods, medical attributes, and its uses in multiple sectors. Scholars have carried out investigations to examine the molecular makeup of black pepper, pinpointing essential constituents accountable for its taste and fragrance. Gas chromatography-mass spectrometry (GC-MS) is one technology that is frequently used in these investigations to give extensive insights into the volatile chemicals present [10]. Black pepper production techniques have been optimised via research on soil needs, irrigation techniques, and insect management. The goal of these studies is to

Development and Validation of an LC-MS/MS Bioanalytical Approach for the Quantitative Analysis of Agomelatine in Human Plasma

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Abstract

An LC-MS/MS Bioanalytical approach has been established to develop a sensitive and quick method for quantifying Agomelatine in human plasma. As an internal standard, Agomelatine d6 was utilized. Ethyl acetate was used to extract agomelatine and internal standard from a plasma sample by liquid-liquid extraction.The upper layer was centrifuged, evaporated, and reconstituted using methanol:5mM ammonium acetate (80:20, v/v) to serve as the mobile phase. Inertsil C18 (50 x 4.6 mm, 5 m) GL Sciences (make) column was used in the method. Multiple reaction monitoring (MS/MS) was used to identify Agomelatine and Agomelatine d6 in human plasma without significant interference from the matrix. The protonated precursor ion of agomelatine ([M+H]+) was detected at m/z 244.80, while the corresponding product ion was detected at m/z 185.10. The protonated precursor ion ([M+H]+) at m/z 250.00 and the matching product ion at m/z 188.10 were both generated by the internal standard. Over a concentration range of 0.0503-8.0055 ng/mL, the analyte calibration curves were linear (R2 0.9956, n=4). After 14 hours on a laboratory bench, 73 hours in an injector, four freeze-thaw cycles, and seven days at -70°C±5°C, Agomelatine was found to be stable in plasma. The developed method was validated in accordance with USFDA guidelines, and theresults were found to be acceptable; this

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A REVIEW ARTICLE ON CHALCONES AS ANTICANCER AGENTS

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Abstract:

Chalcones are the class of compounds which consists of two benzene rings that are connected by alpha, beta unsaturated carbonyl group.(1) They are precursors of flavonoids.(7)

The derivatives of chalcones are important as they contain keto ethylenic group (CO-CH=CH). Since they possess alpha,beta unsaturated carbonyl group, they have a wide range of pharmacological properties like anti proliferative, anti bacterial, anti microbial, anti leishmanial and anti malarial.(2)

This review article mainly focuses on anti cancer activity of chalcones and its derivatives.

Cancer is caused due to uncontrolled growth of abnormal cells and it is a major health problem all over the world. The factors which causes cancer are mainly genetic(5-10%) and environmental(90-95%).(3)

Cancer is considered as second most fatal disease next to cardiovascular diseases which leads to death of many people in the world.(4) Other causes of cancer are use of tobacco(reported 22% of cancer deaths), obesity(10%), poor diet, lack of physical activity or exercise, excessive drinking of alcohol and exposure to ionizing radiation.(5)

Cancer can be identified by signs, symptoms and screening tests which is then investigated by medical imaging and finally confirmed by biopsy.

Sri Vasavi Institute of Pharmaceutical Sciences

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REVIEW ARTICLE

A review on High Performance Liquid Chromatography

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Abstract: High-Performance Liquid Chromatography (HPLC) stands as a pivotal qualitative and quantitative analytical method extensively applied in the estimation of pharmaceutical and biological samples. Recognized as the most versatile and widely employed separation technique, this review article is dedicated to comprehensively surveying the instrumentation aspects of HPLC. Originating from the pioneering work of Csaba Horvath in 1964, the article explores the evolution, advancements, and key components of HPLC instrumentation, shedding light on its instrumental intricacies and technological innovations that have contributed to its enduring significance in analytical sciences. This review aims to offer researchers, scientists, and practitioners a valuable resource for understanding the technological intricacies that underpin the reliability and efficacy of HPLC in analytical endeavors. Through a historical lens and a contemporary viewpoint, the article showcases the instrumental landscape of HPLC, emphasizing its enduring significance and the continuous innovations that shape its trajectory in analytical sciences

Keywords: High-Performance Liquid Chromatography; Qualitative Analysis; Quantitative Analysis; Csaba Horvath; Instrumentation

1. Introduction

Chromatography technique was invented by Dr. Mikhail Semyonovich Tsvet in 1906 later in 1941 Archer John Porter Martin and Richard Laurence Millington Synge discovered Partition Column Chromatography, they have been awarded Nobel prize in 1952 for their invention. Chromatography is a Separation technique in which the mixtures of components are divided into individual components by using Stationary phase and Mobile phase. The HPLC technique was first developed by Csaba Horvath in 1964 [1]. It was the most versatile and widely used separation technique, this technique is used by chemists to separate various biological, oragnic and inorganic compounds, the pressure in HPLC is 5800-7000 psi due to this high pressure it is also called High Pressure Liquid Chromatography [2]. The main aim of this review is to cover the key aspects of HPLC instrumentation, encompassing advancements in column technologies, detector systems, and mobile phase delivery mechanisms.

2. HPLC

2.1. Principle

The Principle of Separation in HPLC is Adsorption, when sample mixture are introduced in column they travel according to their relative affinity, the compound which has more affinity towards Stationary phase travel slower and the compound which has lessaffinity towards stationary phase will travel faster since no two compounds have same relative affinity [3].

2.2. Instrumentation

HPLC instrumentation consists of the following main components as illustrated in the Figure 1:

- Solvent Reservoir
- Pumps
- Injectors
- Columns
- Detectors
- Recorders & Integrators

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A New Stability Indicating RP-HPLC Method for Simultaneous Estimation of Voxilaprevir, Sofosbuvir and Velpatasvir in Bulk and Pharmaceutical Dosage Forms.

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ABSTRACT

A new, simple, rapid and stability-indicating RP-HPLC method was developed and validated for the estimation of Voxilaprevir (VOX), Sofosbuvir (SOF) and Velpatasvir (VEL) in pharmaceutical dosage forms. The HPLC method was developed on SHISEIDO C18 column (250 x 4.6 mm i.d. 5µ) using Acetonitrile : 20 mM acetate buffer (pH 3.0±0.1) in the ratio of 85:15 v/v at 245 nm. The retention times for VOX, SOF and VEL were found to be 2.75, 3.37 and 6.49 min respectively. Linearity was established in the range of 2.5-15 µg/ml, 10-60 µg/ml and 2.5-15 µg/ml for VOX, SOF and VEL respectively. The method was precise with %RSD < 2 for both intraday and interday precision. The accuracy of the method was performed over three levels of concentration and the recovery was in the range of 98-102%. The drugs were individually subjected to forced degradation (thermal, photolytic, hydrolytic, and oxidative stress conditions) studies at different strengths and temperatures. VOX showed highest degradation in acidic and basic condition. SOF and VEL showed highest degradation in peroxide and acidic conditions when compared with basic, photolytic and thermal conditions. The method was successfully applied for quantifying the drugs in marketed dosage forms.

Keywords: Voxilaprevir, Forced degradation, Acetate buffer, RP-HPLC, Sofosbuvir.

common in the United States, and affecting 72% of all chronic HCV patients. Sofosbuvir (Fig 2) is a nucleotide analog inhibitor of HCV NS5B polymerase. It is a direct acting antiviral medication used as part of combination therapy to treat chronic Hepatitis C, an infectious liver disease caused by infection with Hepatitis C Virus (HCV). Velpatasvir (Fig 3) is a Direct-Acting Antiviral (DAA) medication used as part of combination therapy to treat chronic Hepatitis C, an infectious liver disease caused by infection with Hepatitis C Virus (HCV). HCV is a single-stranded RNA virus that is categorized into nine distinct genotypes, with genotype 1 being the most common in the United States, and affecting 72% of all chronic HCV patients. Velpatasvir acts as a defective substrate for NS5A (Non-Structural Protein 5A), a non-enzymatic viral protein that plays a key [1-4].

The present study is to develop a stability indicating RP-HPLC method for VOX, SOF and VEL. The objective of the study is to subject the drugs for acid, base, peroxide, light, thermal degradation and estimate its extent of degradation. A Literature survey reveals that several analytical methods for the estimation of VOX, SOF and VEL based on UV [5-8], HPTLC [9], HPLC [10-23], Stability indicating HPLC [24-30], Related substances [19] Bio-analytical [31], UPLC [32], LC-MS [33,34] were reported.

Although different analytical methods are available, a more economical stability indicating

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A Novel RP-HPLC Method Development and Validation for the Simultaneous Estimation of Drotaverine Hydrochloride and Mefenamic Acid in Pure Drugs and Formulation

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***Abstract*:**
A new, simple and rapid RP-HPLC method was developed and validated for the estimation of Drotavarine hydrhloride (DTA) and Mefenemic acid (MFA) in pure drugs and formulation. The chromatographic separation was achieved on SHISEIDO C18 column (250 x 4.6 mm i.d, 5µ) using Acetonitrile : Methanol (pH 3.0±0.1) in the ratio of 70 : 30 v/v, with a flow rate of 1 ml/min and detection at 303 nm. The retention times for DTA and MFA were found to be 2.1 and 5.7 min respectively.